

Nonsteroidal anti-inflammatory drugs for primary dysmenorrhoea.

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BACKGROUND: Dysmenorrhoea is a common gynaecological complaint consisting of painful cramps accompanying menstruation, which in the absence of any underlying abnormality is known as primary dysmenorrhoea. Research has shown that women with dysmenorrhoea have high levels of prostaglandins, hormones known to cause cramping abdominal pain. Nonsteroidal anti-inflammatory drugs (NSAIDs) are drugs which act by blocking prostaglandin production. **OBJECTIVES:** The purpose of this review is to compare all nonsteroidal anti-inflammatory drugs used in the treatment of primary dysmenorrhoea with placebo, with paracetamol and with each other to evaluate their effectiveness and safety. **SEARCH STRATEGY:** We searched the Cochrane Menstrual Disorders and Subfertility Group trials register (11 April 2003), Cochrane Central Register of Controlled Trials (1st quarter 2003), MEDLINE (1966-April 2003), and EMBASE (1980 - Week 15 2003). Attempts were also made to identify trials from the National Research Register and the Clinical Trials Register. Citation lists of relevant publications, review articles, abstracts of major scientific meetings and included studies were also searched. **SELECTION CRITERIA:** All randomised controlled comparisons of NSAID therapies versus placebo, versus other NSAIDs or versus paracetamol when used to treat primary dysmenorrhoea. **DATA COLLECTION AND ANALYSIS:** Two reviewers independently assessed trials for quality and extracted data, calculating odds ratios for dichotomous outcomes and weighted mean differences for continuous outcomes. Crossover trial data were presented in additional tables and other data were summarised descriptively. **MAIN RESULTS:** In women with dysmenorrhoea, NSAIDs were found significantly more effective for pain relief than placebo (OR 7.91, 95% CI 5.65 to 11.09), though overall adverse effects were also significantly more common (OR 1.52 95% CI 1.09 to 2.12). When NSAIDs were compared with each other or with paracetamol, there was little evidence of the superiority of any individual NSAID with regard to either efficacy or safety. However the available evidence had little power to detect such differences, as most individual comparisons were based on very few small trials, most of which were unsuitable for meta-analysis. **REVIEWER'S CONCLUSIONS:** NSAIDs are an effective treatment for dysmenorrhoea, though women using them need to be aware of the significant risk of adverse effects. There is insufficient evidence to determine which (if any) individual NSAID is the most safe and effective for the treatment of dysmenorrhoea.